

## Multiple Choice (10 marks)

1. What is the total number of colors in RGB color space?
  - (a) 16777216
  - (b) 4096
  - (c) 5000000
  - (d) 32768
  - (e) 1000000
2. The message must be encrypted at the sender site and decrypted at the
  - (a) Data center
  - (b) Sender Site
  - (c) Server site
  - (d) Conferencing
  - (e) Site
3. Let  $X_0, X_1, X_2, \dots$  be drawn i.i.d. from  $p(x)$ , and  $x \in \{1, 2, 3, \dots, 100\}$ . Let  $N$  be the waiting time to the ne
  - (a) 99.0
  - (b) 101.0
  - (c) 125.0
  - (d) 150.0
  - (e) 50.0
4. Given 2 colors whose HSI representations are given as follows: (a)  $(\pi, 0.3, 0.5)$ , (b)  $(0.5\pi, 0.8, 0.3)$ , which color is brighter?
  - (a) Brightness cannot be compared without additional information about the viewing conditions
  - (b) (b) is brighter due to higher hue value
  - (c) Cannot be determined from the given data
  - (d) Brightness is subjective and cannot be determined from HSI values alone
  - (e) (b) is brighter because it has a higher saturation and lower hue value
5. Which of the following is NOT a property of bitmap graphics?
  - (a) They can support millions of colors
  - (b) Realistic lighting and shading can be done.

- (c) Bitmaps can be made transparent
- (d) Fast hardware exists to move blocks of pixels efficiently.
- (e) Bitmap graphics can be created in multiple layers

## True / False (10 marks)

*Answer True or False*

- 6. True or False: WPA2-Enterprise provides the strongest wireless security among commonly used protocols.
- 7. True or False: The function  $O(1)$  represents the smallest asymptotic growth rate among the given options.
- 8. True or False: A heap area is usually not included in a subroutine's activation record frame in stack-based programming languages.
- 9. True or False: The capacity of a cascade of  $n$  identical independent binary symmetric channels with error probability  $p$  approaches 0.0 as  $n$  goes to infinity.
- 10. True or False: Overfitting refers to a model that performs well on the training data but fails to generalize to new data.

## Long Form Response (20 marks)

*Answer each question with a clear and complete explanation.*

- 11. Write a function to find the list of lists with maximum length.
- 12. Write a function to separate and print the numbers and their position of a given string.

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